DYANEL LANCEA E. BALANDRA

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PROFESSIONAL SUMMARY

I am a highly motivated Undergraduate student pursuing a course in Fintech. I am eager to transition my technical skills into the professional Fintech and RegTech space. Currently seeking a hands-on volunteer opportunity to apply academic knowledge, and expand expertise in real-world Fintech industry.

EDUCATION

BSc (Hons) in Applied Computing (specialisation in Fintech) **Sep 2025 - Current**
Singapore Institute of Technology
Year 1, Trimester 1
Relevant Courses: Programming Fundamentals, Introduction to Computer Systems

Diploma in Computer Engineering **Apr 2022 - Mar 2025**
Temasek Polytechnic
Cumulative GPA: 3.37/4.0, Temasek Polytechnic Scholarship Engineering (2022-2024)
Relevant Courses: Full-Stack Development, AI & Machine Learning, Internet of Things (IoT), Object Oriented Programming

WORK EXPERIENCE

Zenitant **July 2025 - Current**
Freelance, STEM Trainer

- Managed and taught students aged 10–14 by designing and delivering coding-based lessons on IoT, robotics, drones, and STEM, resulting in enhanced technical proficiency and increased interest in advanced STEM topics.

EP-Tec Solutions **Apr 2024 - Aug 2024**
Intern, EP-Edu dept.

- Designed and curated comprehensive coding curriculum for students aged 10–14, successfully implementing programs in IoT, robotics, and drones directly broadening the department's technical offerings.
- Elevated the technical expertise of the instructional team by training them on specialized equipment (e.g., mBot2), ensuring consistent and high-quality lesson delivery across all technology courses.

PROJECTS

Silver Companion SG **(3 Days) Dec 2025**
Hackrift Hackathon 2025 | Finalist

- Engineered an agentic AI ecosystem designed to assist Singaporean elderly, consisting of a central orchestrator, wellness, and safety agents.
- Implemented Singlish-optimized voice interaction capabilities to ensure high accessibility for local seniors, bridging the digital divide through natural vernacular communication.
- Designed a resilient response framework to handle real-time emergency scenarios and routine health check-ins, prioritizing user safety and mental well-being monitoring.

Tech: FastAPI, Sea-Lion LLM, Whisper (STT), Supabase, Twilio API

PROJECTS

Flood Drone Monitoring System **Oct 2025 - Feb 2025**

Major Project, Temasek Polytechnic

- Developed an IoT+AI drone system (ESP32, Raspberry Pi, DJI Tello) that collects temperature, humidity, and aerial imagery to detect potential flood risks in inaccessible areas.
- Built a full TIG data pipeline and AI image-analysis workflow using Google Vertex AI to identify high water levels or blocked drains, with results stored in InfluxDB and visualised in Grafana.
- Implemented real-time Telegram alerts and a monitoring dashboard, and designed the system prototype housing in TinkerCad.

Tech: ESP32, Raspberry Pi 4, InfluxDB, Grafana, Telegraf, Vertex AI, Telegram Bot API

ACTIVITIES

SIT Breakdance Club **Jan 2026 - Current**

Main-Committee | Head of Publicity

- Regularly update club's social accounts (Instagram, TikTok).
- Monitor lesson recordings.

Engineering Studies Club (ESC) **Aug 2022 - Mar 2025**

Sub-Committee | Facilitator

- Coordinate and manage events under the School of Engineering in Temasek Polytechnic.
- Facilitate event activities and efficiency.
- Monitor participants and their concerns.

Tchoukball Temasek Polytechnic **Apr 2023 - Mar 2025**

- Participated in POL-ITE 2023 (Silver).
- Participated in POL-ITE 2024.

TECHNICAL SKILLS

Software tools:

- Visual Studio & Visual Studio Code
- Supabase
- Postman
- JupyterLab
- SQL Database Management Software
- MakeCode Blocks Playground

Microcontrollers & Others:

- Raspberry Pi
- Arduino
- Tello Drone
- Microbit & mBots

Languages:

- Python, C
- HTML/CSS/JS
- TypeScript
- Java